

Manikantan (Mani) Srinivasan

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EDUCATION

Northeastern University, Boston, Massachusetts

Sep 2023 – Aug 2025

Khoury College of Computer Sciences

(expected)

Master of Science in Artificial Intelligence - GPA: 3.5

Coursework: NLP, AI in Human Computer Interaction, LLM, Machine Learning, Data Science, Unsupervised ML

Teaching Assistant for CS6220 Data Mining – teaching graduate-level data mining techniques.

UPES, Dehradun, Uttarakhand (India)

Aug 2019 – May 2023

Bachelor of Technology in Computer Science (specialization in AI&ML) – GPA: 8.3

Coursework: Machine Learning, Algorithms, Deep Learning, Database Management, Software Engineering

SKILLS

Languages:

Python, C++, Java, C, R, HTML, CSS, Bash, MATLAB, Dart, Swift

Software Development Tools

Git, GitLab, Jira, Docker, Kubernetes, Xcode, Tableau, Snowflake

Cloud and Databases

GCP, AWS EC2, SageMaker, S3, ArangoDB, MySQL, MongoDB, Neo4j, Redis, Flutter,

Libraries and Frameworks:

PyTorch, HuggingFace, TensorFlow, Django, LangChain, Pandas, Sci-kit Learn, Matplotlib

Domain Skills

Statistical Modelling, Deep Learning, LLM Fine Tuning, RAG, AI Agents, Data Analytics

PROFESSIONAL EXPERIENCE

SignoFi, Raleigh, USA

Sep 2024 – Jan 2025

AI Research Engineer Intern

- Pioneered an Agentic AI application leveraging Llama 70B and LangChain to retrieve balance sheet data from NoSQL databases, enabling interactive queries and computational analysis for over 50,000 public companies.
- Leveraged RAG and Finetuning with adaptive chunking strategies to ground Llama 70B and 90B, boosting the Faithfulness score to 0.84 (84%) and enhancing contextual precision, recall, and relevancy (92%) in factual question answering.
- Architected a custom workflow on Google Cloud Platform using virtual machines, Vertex AI, GCP Buckets, and multi-core GPUs to optimize operations at a portfolio management firm, achieving up to 20% in cost savings.

KEK, Tsukuba, Ibaraki (Japan)

Oct 2022 – May 2023

Researcher – Distributed Computing (International Research Lab)

- Collaborated on the Prestigious [BELLE-2](#) particle acceleration experiment as a student researcher. The collaboration involved 5+ world famous laboratories like Brookhaven, CERN, KEK and 127+ institutions like Carnegie Mellon.
- Engineered innovative software tools that empowered physicists to execute experiments on over 10 PB of data across global data centers.
- Streamlined 50+ issues on GitLab and JIRA by overhauling complex codebases and partnering with engineers worldwide.

Jio, Mumbai, Maharashtra (India)

Jun 2022 – Aug 2022

Software Engineer Intern – (Center of Excellence- AI & ML)

- Engineered the [BrainOS](#) platform integrating 7+ verticals for Reliance Industries, utilizing knowledge graphs and predictive ML across products like JioTV, Hydrocarbons, and JioMart, achieving 86% accuracy and 0.79 precision.
- Developed robust knowledge graphs by preprocessing raw data and implemented 100+ CRUD API functions using gRPC to efficiently query extensive NoSQL databases, utilizing debugging tools to analyze and refine 1000+ functions.
- Facilitated the development of a beta version of user-user collaborative filtering for product recommendations using Neo4j and custom graph functions as part of a cross-functional effort, helping drive a 10% revenue boost for Jio Platforms.

PROJECTS

[BlindSight](#): AI-Powered Voice-Driven File Navigation for the Visually Impaired

Aug 2024 – Dec 2024

- Developed an **AI-powered voice navigation system**, enabling visually impaired users to perform file management and document editing tasks with 95% speech recognition accuracy using Open-AI's C++ implementation of **Whisper**.
- Incorporated **LLaMA-70B** with tailored in-context learning (ICL) to deliver task-specific capabilities, such as document summarization and advanced text editing, achieving 85% accuracy in real-world testing scenarios.
- Designed a multi-threaded architecture that reduced task execution latency by 30%.

DCGAN vs VAE: Comparative Analysis of prominent Image Generation Techniques

Apr 2024 – May 2024

- Implemented and evaluated **DCGANs** and Variational Autoencoders (**VAEs**) using **PyTorch** to generate and reconstruct images from CIFAR-10, achieving an **IS of 5.48** and an **FID of 58.4**, demonstrating superior generative performance of DCGAN over VAE. - [Report](#)

ACHIEVEMENTS

- Selected (from a pool of 500 students) for the prestigious BELLE-2 Particle Accelerator experiment, a Nobel Prize-winning endeavor. - <https://www.belle2.org>